PULSE: Data, Database & AI: Full Report

Everything the data/AI workstream built for the PULSE hypertension-monitoring prototype: how the parts relate, every table, every dataset, and every model with its training data, test data and results.

30 October 2026 · Data/AI workstream · Branch `AI` of `github.com/Ayoub-Meziani/Pulse` · Cloud project `pulse` (Supabase, Paris). Prototype, not a medical device. Patient data is **fictional** and labelled `simulated`. Real public datasets were used only for statistics, calibration and tests.

8 / 8 migrations, cloud in sync	**24** tables, all with access rules (52 policies)	**14** demo accounts (1 doctor, 1 admin, 12 patients)	**730** blood-pressure readings in the cloud
20,832 simulated ring samples (12 patients × 28 days)	**5** models: 3 in use, 1 experiment, 1 not shown	**3** real public datasets (NHANES, Mendeley, PhysioNet)	**86** automated tests (database, data, ML)

Status labels: LIVE in the cloud database · BUILT code exists and is tested · EXPERIMENT evaluated, not used · TO BUILD another owner's next step · MISSING nobody has it yet · TO VERIFY check before use

Contents

1. Where we are
2. System map: how everything connects
3. What relates to what
4. Database
5. Data: synthetic, demo and real
6. AI models: training, test data, results
7. Live ring monitoring and the weekly doctor loop
8. How the jobs run (inference contract)
9. Tests and quality
10. Commands
11. Safety, ethics and limits
12. Next steps
13. Sources

1. Where we are

Part	Status	Details
Cloud database `pulse`	LIVE	Migrations M1–M8 applied and in sync (checked 30 Oct). 24 tables, row-level security (RLS) on every table, 52 policies, nothing granted to logged-out users, public sign-up off.
Demo data in the cloud	LIVE	14 accounts; 730 readings (727 valid, 2 needs_review, 1 invalid, on purpose); 316 lifestyle logs; 154 dose events; 11 care plans; 24 body measurements; 24 devices; 20,832 ring samples for all 12 patients. All labelled `simulated`. Password only in the local, git-ignored `.env`.
ML outputs in the cloud	LIVE	Jobs ran on the cloud on 30 Oct (`scripts\run_cloud_jobs.ps1`): 10 next-day BP estimates + 1 `insufficient_history`; 33 ring flags (22 outside reference, 10 personal, 1 multivariate); 11 weekly reports, all with the classifier probability; 43 proposed bands waiting for the doctor.
Alerts in the cloud	TO BUILD	0 rows. The alert rules live in the backend (FastAPI, teammate), which still runs on SQLite. The tables and the episode logic (M3) are ready.
5 models (`ml/`, Python)	BUILT	`mbp-v1.1` next-day BP (served), `ring-v1.1` ring flags, `wcls-v1.1` weekly classifier, `mbp-v2.1` (experiment), `wrisk-v1` on real NHANES people (evaluated, not shown). Section 6.
Real public data	BUILT	NHANES 2017–2020 (CDC), Mendeley home-BP series, PhysioNet Apnea-ECG. Downloaded to git-ignored `data/external/` with a manifest, never committed or uploaded. Section 5.6.

Device (cuff / ring)	MISSING	No physical device yet. Tables for cuff readings and ring samples are live; ring data is simulated from published values.
Real patient data	MISSING	None, on purpose. No open multi-patient home-BP dataset exists, so real accuracy needs consented PULSE data later.

2. System map: how everything connects

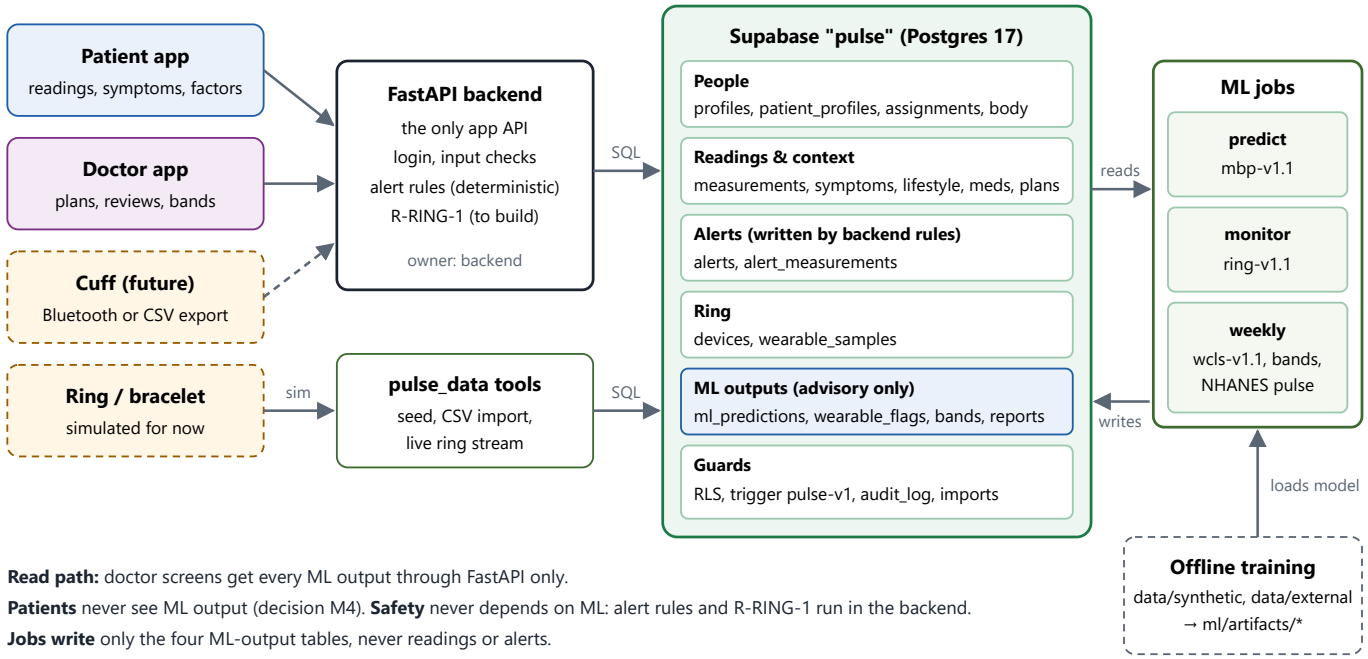


Figure 1. Data enters through FastAPI or the data tools, lands in Supabase, and the three ML jobs read it and write back only advisory outputs. Dashed = not built yet or simulated.

Component	Talks to	Credential it uses	Where that credential lives
Frontend (patient, doctor)	FastAPI only (team decision A2)	the user's own login token	nothing stored
FastAPI backend	Postgres; Supabase Auth (JWT checked with JWKS, ES256)	database URL, Supabase URL, anon key	server `.env` / hosting secrets
ML jobs, CSV import, seed	Postgres directly	`DATABASE_URL`	data/AI machine `.env` (git-ignored)
Migrations	Supabase CLI `db push`	project owner's CLI login	never shared

Never: the service-role key or the database password in the frontend, git, chat or screenshots. The GitHub repo is **public**.

3. What relates to what

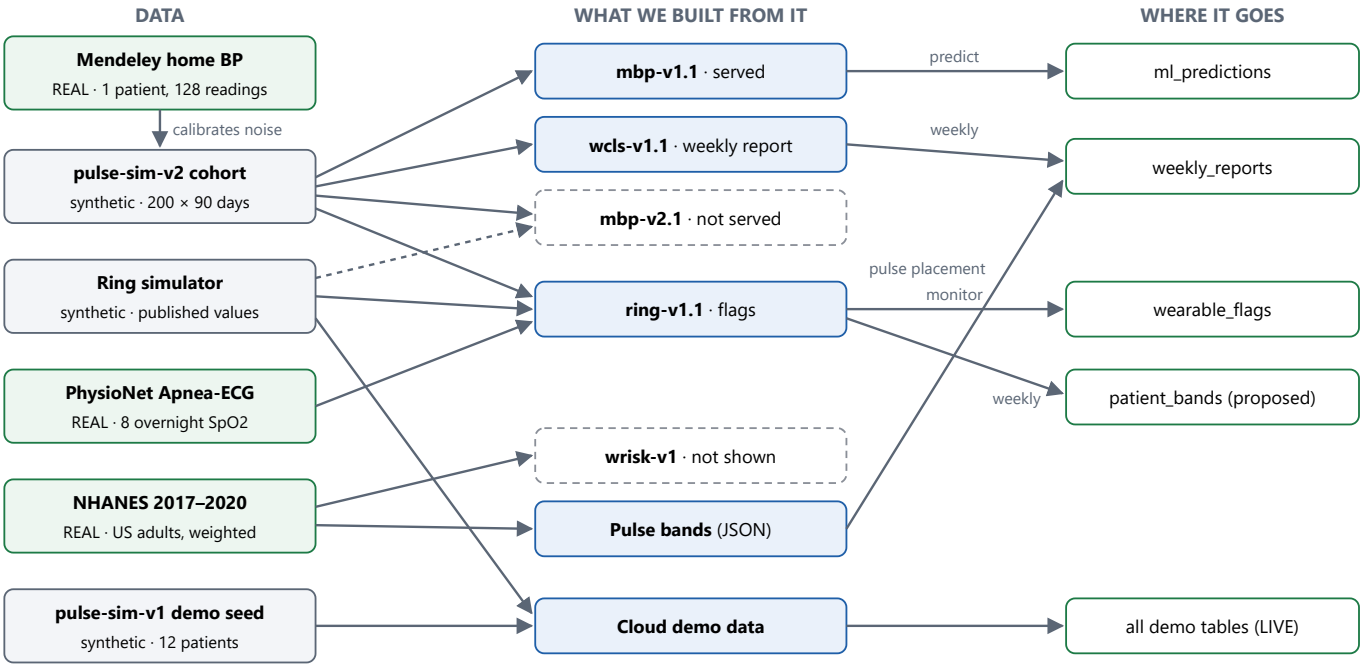


Figure 2. Every dataset, what it produced, and the table each result lands in. Mendeley calibrates the v2 simulator; Apnea-ECG tests the ring SpO2 rule; NHANES gives the pulse bands and wrisk-v1. Green = real public data; dashed = evaluated but not used in the product.

Component	Code	Reads	Writes	Who sees it	Status
Validation trigger `pulse-v1`	migration M1	each new reading	validation_status, quality_flags	everyone (on the reading)	LIVE
Access rules (RLS)	M2, M3, M6, M7	caller's role + assignments	-	-	LIVE
Demo seed	`pulse_data/simulate.py seed`	STEPS-DZ shares, ring values	all demo tables	team, demo	LIVE
Cloud ring refresh	`simulate ring-demo-sql`, `scripts/cloud_update.ps1`	ring simulator	wearable_samples (simulated only)	doctor	LIVE
CSV import	`pulse_data/ingest.py`	device CSV export	measurements, ingestion_batches	patient, doctor	BUILT
Live ring stream	`pulse_data/stream.py`	ring simulator	wearable_samples	doctor (via flags)	BUILT (local)
Next-day BP	`ml/pulse_ml/predict.py` + `mbp-v1.1`	valid readings, assignments	ml_predictions	doctor only	BUILT
Ring monitor	`monitor.py`, `ring.py` + `ring-v1.1`	wearable_samples, active bands, conditions, confirmed flags	wearable_flags	doctor only	BUILT
Weekly report	`weekly.py` + `wcls-v1.1`, `bands.py`, NHANES bands	readings, care plans, ring nights, flags, predictions, profile	weekly_reports, patient_bands (proposed)	doctor only	BUILT
Band approval	`private.approve_band` (M7)	proposed band	patient_bands (active), audit_log	doctor acts	LIVE (API TO BUILD)
Feedback gate	`feedback.py`	reviewed flags	git-ignored labels file	data/AI	BUILT
Medical references	`reference.py`	-	-	used by monitor, weekly, bands	REVIEWED 30 Oct

Real-data tools	`real_data.py`, `real_bands.py`, `wrisk.py`, `real_tests.py`	data/external (git-ignored)	reference_nhanes.json, metadata	-	BUILT
Alert rules	backend `rules.py`	readings, symptoms, care plan	alerts, alert_measurements	patient + doctor	TO BUILD on Supabase

4. Database

4.1 Migrations (one append-only history in `supabase/migrations/`)

#	File	What it added, and why	Cloud
M1	`20261029120000_core_schema`	17 tables; validation trigger `pulse-v1`; audit triggers; role checks; profile row on sign-up. The base of every feature.	LIVE
M2	`20261029120100_access_policies`	RLS on every table; `anon` gets nothing; per-role write policies; column grants (e.g. a patient cannot change their role).	LIVE
M3	`20261030120000_backend_alignment`	Backend contract: `above_target_trend` alert class, `alerts.persistent`, table `alert_measurements` (same-patient keys), `measurements.rested_5_min`, empty symptom answers allowed, doctor notes hidden from patients.	LIVE
M4	`20261030120100_ml_predictions`	Table `ml_predictions`: values present only when status is `ok`, low ≤ high; doctor/admin read only; no client writes.	LIVE
M5	`20261030130000_patient_birth_year`	`patient_profiles.birth_year` (year only, data minimisation). Age is the strongest Algerian hypertension factor.	LIVE
M6	`20261030140000_algerian_factors_wearables`	Sex, smoking, condition codes, bread / salt habits, language, fasting logs; tables `body_measurements` and `wearable_samples` (unit and plausible range checked per metric).	LIVE
M7	`20261030150000_ml_monitoring`	Tables `wearable_flags`, `patient_bands`, `weekly_reports`; function `private.approve_band`; audit on all three; index for the live monitor. Built exactly as the backend's ML review asked.	LIVE
M8	`20261030160000_chronic_conditions`	Condition codes add copd, asthma, obstructive_sleep_apnea, atrial_fibrillation, heart_failure, thyroid_disease (they change how ring values are judged). Applied 30 Oct after team approval.	LIVE

4.2 How the tables relate

- **One person = one login** (`auth.users`) = one `profiles` row (role patient / doctor / admin). Patients have one `patient_profiles` row.
- **Every patient-owned table** has `patient_id → profiles.id`. A doctor's access is never stored per row: it is computed from an **active** `doctor_patient_assignments` row by the helpers `private.is_assigned_doctor()` and `private.is_admin()`.
- **Same-patient keys:** a reading can only use the patient's own device; an alert can only attach the same patient's readings (`alert_measurements`, composite keys); ring samples can only come from the patient's own device.
- **Chains:** `measurements → alerts → alert_measurements`; `devices → wearable_samples → (monitor) → wearable_flags → band_id → patient_bands`; `patient_bands.supersedes_id` and `care_plans.supersedes_id` keep history instead of overwriting.

- **History, never silent deletes:** readings are append-only; wrong readings stay, flagged; changes to sensitive tables are copied to `audit_log` (old row, new row, actor).

4.3 Every table and field (24 tables)

Every table also has `id` (except link tables) and `created_at`, filled automatically. **Blue** = patient, **purple** = doctor, **green** = system / device / admin. Column counts are from the cloud.

People and access

Table	Purpose	Patient fills	Doctor fills	System / admin
profiles (7)	One row per account	display_name, timezone (default Africa/Algiers), preferred_language (fr / ar)	same	id = login id, role (admin only), updated_at
patient_profiles (18)	Background and risk factors	birth_year, sex, smoking_status, condition_codes, relevant_conditions, bread_portions_per_day, adds_salt_at_table, salty_processed_food, allergies, medication_context, regular_clinic, emergency_preferences, consent_location, data_consent_at	(reads)	user_id, regular_doctor_id, updated_at
body_measurements (9)	Weight, height, waist	measured_at, weight_kg, height_cm, waist_cm (patient_reported)	same (clinician)	created_by; no edits or deletes
doctor_patient_assignments (7)	Who follows whom; drives all doctor access	(reads)	(reads)	doctor_id, patient_id, status active / ended, ended_at
audit_log (8)	Who changed what	-	-	table_name, row_id, action, actor, changed_at, old_data, new_data (admin reads)

Readings, devices, ring

Table	Purpose	Patient fills	Device / import	System
measurements (24)	Every BP reading (append-only)	systolic, diastolic, pulse, measured_at, measured_tz, body_position, arm, context, rested_5_min	device_id, ingestion_batch_id, external_id, method, raw_payload, device flags	source, provenance, unit mmHg, validation_status, quality_flags, validation_policy_version, created_by
devices (9)	Cuffs, rings, phone apps	device_type, manufacturer, model, external_device_id, label, bp_capability	-	patient_id
wearable_samples (10)	Ring / bracelet values	-	device_id, metric (heart_rate bpm, hrv_rmssd ms, spo2 percent, skin_temperature celsius), value, measured_at, external_id	source (device_sync / csv_import / simulated); unit + range checked; one row per (device, external_id)

ingestion_batches (13)	One row per import or sync	-	source, device_id, file_name, file_sha256 (same file never twice)	row counts, rejections (row + reason, no health values)

Patient-reported context

Table	Purpose	Patient fills	System
symptom_reports (8)	Symptoms, judged independently of BP	symptoms (none, or chest_pain, shortness_of_breath, back_pain, weakness, numbness, vision_change, difficulty_speaking, severe_headache, confusion, dizziness, nausea, palpitations, other), onset_at, notes	reported_at, source
lifestyle_logs (17)	Meals, activity, sleep, stress, fasting	category, started_at, ended_at, description; meal_type, sodium_mg + basis; activity_type, intensity, steps; sleep_quality 1–5; stress_level 1–5	source, updated_at
medication_adherence_events (10)	Each dose taken / missed / skipped	medication_id, status, scheduled_for, taken_at, note	source, reported_at
journal_entries (8)	Notes / questions for the doctor	category, entry_text	review status, reviewed_by / at (doctor)
trusted_contacts (7)	Emergency contacts	name, relationship, phone, consented_at, active	-
emergency_events (12)	Emergency screen history	trigger_reason, alert_id, symptoms_snapshot, started / ended_at, actions_attempted ("tapped", never "called"), reported_outcome, location (only with consent)	-

Doctor-authored

Table	Purpose	Doctor fills	Notes
care_plans (17)	Targets and monitoring plan	target_systolic / diastolic, readings per day, instructions, lifestyle and follow-up text, effective_from / to, review_date, status, supersedes_id	One active plan per patient; default target 140/90 if none
medications (12)	Prescriptions and own meds	name, dose_text, schedule_text, active, started_on, ended_on	PULSE never computes or changes a dose
doctor_notes (8)	Private note + advice	note, advice, follow_up_at	Patients get only advice and follow-up date, through the API

Alerts and ML outputs (system)

Table	Purpose	Fields	Who can act
alerts (20)	Rule results	alert_class, rule_code, rule_version, reason, actions, persistent, measurement_id, symptom_report_id, trigger_source, status + timestamps, resolution_note	patient acknowledges; doctor reviews, resolves
alert_measurements (4)	Readings of one alert episode	alert_id, measurement_id, patient_id, attached_at	read only
ml_predictions (16)	Next-day BP estimate	target_date, target, status (ok / insufficient_history), predicted_systolic / diastolic, 4 interval bounds, model_version, trained_on, features_until, n_readings_used	doctor reads only

wearable_flags (17)	Advisory ring flags	metric (+ multivariate), kind (outside_reference / personal_anomaly / multivariate_anomaly), window_kind (night / live_10min), window_start / end, value, score, reference_basis, band_id, model_version, review_status (open / confirmed / dismissed), reviewed_by / at, review_note	doctor updates review fields only
patient_bands (14)	Personal ring normal range	metric, statistic (night_median / night_min / live_10min_median), low, high, basis (reference / model_proposal / doctor), status (proposed → active → superseded, or rejected), proposed_model_version, approved_by / at, effective_from, supersedes_id	doctor approves via `private.approve_band` or rejects
weekly_reports (14)	Weekly summary per patient	week_start (local Monday), home_bp_category, control_status, next_week_above_target_prob, model_versions, summary (numbers only, ≤ 16 KB), status ready / reviewed, reviewed_by / at, doctor_comment	doctor reviews and comments

4.4 Rules inside the database

Rule	What it does
Validation `pulse-v1` (trigger, every reading)	**invalid** (kept, not used): systolic ≤ diastolic, systolic outside 50–300, diastolic outside 25–200, pulse outside 25–250, more than 5 minutes in the future. **needs_review**: systolic outside 70–250 or diastolic outside 40–150, an estimate (not a cuff), the same values within 2 minutes, a device-reported problem. **valid**: everything else. A very high reading (186/118) is valid data and an alert matter, not a data problem.
Checks (constraints)	Units fixed per metric; plausible ranges; ML predictions have values only if `ok`; a band's low < high, and an active band needs an approver; a flag's review time is set exactly when it's reviewed; report weeks start on Monday; summaries stay small and numeric.
`private.approve_band(band, doctor, low, high)`	One transaction: the active band for that metric becomes `superseded` and the proposed one becomes `active`. If the doctor edits low / high, the basis becomes `doctor`. Checks that the doctor is actively assigned and is the caller. It lives in the private schema, so it is not a public endpoint.
Audit triggers	Inserts and updates on alerts, care plans, bands, flags, reports and other sensitive tables are copied to `audit_log`.
One active row	One active care plan per patient; one active band per (patient, metric, statistic); one flag per (patient, metric, kind, window start); one rule firing per reading / symptom report.

4.5 Who can see what (RLS + grants)

Data	Patient	Assigned doctor	Admin	Logged out
Own profile, readings, context, devices	read + insert own	read	read	nothing
Care plan, medications	read	read + write	read	nothing
Doctor notes	no direct read (API shows advice only)	read + write	read	nothing
Alerts	read, acknowledge	read, review, resolve	read	nothing
ML outputs (predictions, flags, bands, reports)	**nothing** (decision M4)	read; review fields only	read	nothing
Ring samples	read own	read	read	nothing
Audit log	-	-	read	nothing

A doctor who is not assigned sees nothing about a patient. ML jobs and the seed use the privileged database connection; no app user can write ML outputs. The live REST API refuses logged-out requests (error 42501, tested).

5. Data: synthetic, demo and real

Dataset	Kind	Size	Used for
`pulse-sim-v1` demo seed	synthetic	12 patients × 30 days, 730 readings; ring 28 days	Cloud demo accounts; kept byte-stable by fingerprint tests
`pulse-sim-v2` ML cohort	synthetic	200 patients × 90 days, 36,890 readings; 1,005,144 ring samples; 67 labelled ring episodes	Training and testing `mbp-v1.1`, `ring-v1.1`, `wcls-v1.1`, `mbp-v2.1`
NHANES 2017–March 2020	real	7,296 adults with pulse bands; 2,309 treated adults for `wrisk-v1`	Population pulse / BP bands; real classifier test
Mendeley pjcv33hpp7	real	1 patient, 128 home readings over 384 days	Calibrating the simulator's day-to-day variation
PhysioNet Apnea-ECG	real	8 overnight SpO2 recordings	Testing the ring SpO2 rule on real nights

5.1 How the BP simulator works (`pulse_data/simulate.py`)

Each synthetic patient has a baseline systolic (cohort: uniform 110–165 mmHg), a diastolic of about 0.62 × systolic, a slow trend, a chance of skipping a session, and (60 % of them) a medication with a chance of missing doses. Every day the generator draws the day's context, then a hidden **BP effect**:

effect(day) = latent drift + day noise + trend × day + **6** if a dose was missed yesterday + **1.2** × (sodium − 3,200 mg) / 1,000 − **2** if activity ≥ 30 min + **2** if sleep < 6 h
latent drift = φ × yesterday's drift + Normal(0, σ) · **reading**: systolic = baseline + effect + 4 (morning surge) + Normal(0, 5); diastolic = baseline + 0.55 × effect + 2 + Normal(0, 3.5), at least 15 below systolic; pulse ≈ Normal(72, 6), +3 in the morning.
Sessions around 07:00 and 20:00 (−40 to +60 min), 1–2 readings each; 0.3 % of readings have systolic and diastolic swapped on purpose, to test data cleaning.

The effects of missed doses, salt, activity and sleep are **invented** for testing the pipeline; they are not clinical estimates.

5.2 Calibrated on real home BP: v1 vs v2

Systolic, 1 reading every 3 days	SD (mmHg)	change between readings	lag-1 correlation	Noise setting
Real patient (Mendeley, n = 128)	11.7	9.97	0.40	-
`pulse-sim-v1` (first generator)	6.7	6.45	0.24	φ 0.70, σ 3.0, no day noise
`pulse-sim-v2` (median of 111 patients)	11.8	10.04	0.41	φ 0.95, σ 3.6, day noise SD 6.0

The first generator was about half as variable as real home BP, which made the first model look too good (MAE 5.2). v2 was chosen from a small grid to match the real series; all models were retrained (`*.1`). Diastolic v2 6.8 / 5.8 / 0.41 vs real 6.4 / 5.1 / 0.37. One real patient gives a realistic size, not a population fit. The demo seed stays v1, so the cloud accounts did not change.

5.3 How the ring simulator works (values from published studies)

Simulated value	Based on
Night resting HR ~ 65 ± 7 bpm between people, ~3 bpm night to night; every 10 min at night, hourly by day	Quer et al., PLOS ONE 2020: 92,457 wearable users, mean 65 bpm
Night SpO2 ~ 96.5 %, 6 samples per night; one 3–5 % dip on half the nights	Gries & Brooks, Chest 1996: lowest sleep saturation 90.4 % in healthy people. This is why PULSE uses the night **median**
COPD: personal SpO2 4 points lower	BTS 2017: target 88–92 % if at risk; MENA COPD 5.2 % men / 1.8 % women (BREATHE)
Sleep apnea: 2–4 dips of 4–9 % per night, night HR +3	Desaturations ≥ 4 % linked to resistant hypertension (cohort of 284). The 30 % rate is an assumption

Type 2 diabetes: HRV (RMSSD) 0.92 SD lower	HRV in T2DM meta-analysis, PLoS ONE 2018
Atrial fibrillation: noisy HR, meaningless HRV	Irregular pulse on wrist PPG (Apple Heart Study 2019); rates are assumptions
HRV ~ 34 ± 9 ms, skin temperature ~ 34 °C	assumptions (personal baseline only)

Invented coupling (for testing only): night HR +0.25 bpm and HRV −0.3 ms per mmHg of the hidden BP effect. **Injected episodes** in the ML cohort (30 % of patients, 1–3 days, after day 35): `hr_rise` +15 bpm (18), `spo2_dip` −5 % (22), `hrv_drop` −40 % (27). 10 % of days have no ring data (ring not worn). Cohort conditions: sleep apnea 56, diabetes 32, COPD 8, atrial fibrillation 5 of 200. Ring values use a separate random generator, so the BP data never changes (tested).

5.4 The demo patients (fictional; risk factors from real Algerian statistics)

Patient	Story	Sex, age	Smoking	Condition codes	BMI	Waist
Amina	controlled	F, 58	never	hypertension	26.5	86
Karim	rising trend, gaps; seeded night HR rise	M, 65	current	hypertension, diabetes, dyslipidemia, sleep apnea	31.2	104
Nadia	severe event + chest pain	F, 69	never	hypertension, kidney disease	33.0	99
Yacine	sparse, **not assigned**	M, 47	former	hypertension	24.0	90
Fatima	cohort; COPD story	F, 47	never	hypertension, COPD	24.8	73
Khadija	cohort	F, 31	never	hypertension	26.9	89
Samir	cohort	M, 43	current	hypertension, diabetes, dyslipidemia, sleep apnea	22.7	99
Rachid	cohort	M, 44	never	hypertension	19.5	91
Mourad	cohort	M, 66	never	hypertension, dyslipidemia	19.1	96
Djamel	cohort	M, 42	never	hypertension, dyslipidemia	24.2	85
Houria	cohort; atrial fibrillation story	F, 67	never	hypertension, dyslipidemia, atrial fibrillation	26.5	88
Hocine	cohort	M, 55	never	hypertension, diabetes	17.9	85

Plus **Dr. Leila H.** (doctor, assigned to 11 patients) and an admin. The 8 cohort patients' sex, age band, smoking, diabetes, cholesterol, BMI band and abdominal obesity are drawn from the Algerian STEPS 2016–2017 shares (Moussouni et al. 2022): 55 % women among hypertensives; age 30–44 / 45–59 / 60–69 = 39.3 / 27.7 / 11.6 %; smoking 11.1 % current; diabetes 17.5 %; high cholesterol 29.6 %; obese 24.9 %; abdominal obesity 67.8 % (waist > 94 cm men, > 80 cm women). Height, salt answers, language and ring values are labelled assumptions. With 8 people the drawn shares can't match the published ones exactly.

5.5 Do the fields cover the Algerian blood-pressure factors?

Factor (STEPS 2016–2017)	Where it lives	Status
Age (strongest: odds ratio 12.3 for 60–69 vs 18–29)	`patient_profiles.birth_year`	LIVE
Sex (34.2 % women vs 28.3 % men)	`patient_profiles.sex`	LIVE
Obesity, abdominal obesity (OR 1.7 each)	`body_measurements` : weight, height, waist; BMI computed	LIVE
Cholesterol (OR 2.0), diabetes, kidney, heart, stroke, chronic illness	`condition_codes` (13 codes after M8)	LIVE
Salt / bread	`bread_portions_per_day` , `adds_salt_at_table` , `salty_processed_food` , meal `sodium_mg`	LIVE

Smoking, activity, sleep, stress	`smoking_status` ; `lifestyle_logs`	LIVE
Ramadan fasting	`lifestyle_logs` category `fasting` (medication timing stays the doctor's decision)	LIVE
Language, local time	`preferred_language` fr / ar; timezone Africa/Algiers	LIVE
Emergency number	backend config: **14** (Protection civile), decided 30 Oct; never dialled in tests or the demo	DECIDED

5.6 Real public datasets (official sources only, no Kaggle)

Source	What it is	Terms	What we used	Limits
NHANES 2017–March 2020 (CDC / NCHS)	Real exams: 3 oscillometric BP and pulse readings (Omron HEM-907XL, 5 min seated), BMI, waist, smoking, diagnoses, survey weights	NCHS agreement: statistics only, no re-identification, no linkage	Weighted pulse / BP percentiles by sex × age; `wrisk-v1`	US, not Algeria; exam BP, not home BP; self-reported diagnoses; one visit
Mendeley pjcv33hpp7 (Dvornik et al. 2017)	128 home readings (mean of 3 trials) of 1 treated 67-year-old man, every 3 days for 384 days	CC BY-NC 3.0: non-commercial, credit	Calibrating `pulse-sim-v2`	1 patient; no clock times; drop it if PULSE becomes commercial
PhysioNet Apnea-ECG 1.0.0 (Penzel et al. 2000)	8 real overnight SpO2 recordings: 4 apnea, 1 borderline, 3 control	ODC Attribution 1.0	Testing the ring SpO2 rule	8 nights; sleep-lab oximeter, not a ring

How they're handled: `python -m pulse_ml.real_data fetch all` downloads into git-ignored `data/external/<source>/` and writes a `MANIFEST.json` (URL, licence, date, sha256). A changed file is never silently overwritten. Only aggregated numbers and trained models are committed, labelled `real:<source>`. Kaggle was rejected: it needs an account token, and the popular BP sets have unclear licences or origins.

5.7 Data rules that hold everywhere

- Every row says where it came from: `source` = manual / device_sync / csv_import / **simulated**; ML artefacts say `synthetic:pulse-sim-v2` or `real:nhanes`.
- Only **valid** readings feed averages, trends and ML; needs_review and invalid readings are counted, not used.
- No health values in logs or error messages; jobs print counts only.
- Raw real data never leaves the laptop; the demo password and the database URL live only in `.env`.

6. AI models: training, test data, results

Model	Question it answers	Algorithm	Trained on	Writes	Status
`mbp-v1.1`	What will tomorrow's morning BP be?	ridge regression	synthetic v2	ml_predictions	SERVED
`ring-v1.1`	Is tonight's ring data unusual for this person?	robust z + IsolationForest + references	synthetic ring	wearable_flags	IN USE (flags)
`wcls-v1.1`	Will next week's home mean be above target?	logistic regression	synthetic v2	weekly_reports	IN USE (report)
`mbp-v2.1`	Does ring data improve next-day BP?	ridge regression	synthetic v2 + ring	-	EXPERIMENT
`wrisk-v1`	Is a treated adult's BP uncontrolled (≥ 140/90)?	logistic regression	**real** NHANES	-	NOT SHOWN

All models are Python (pandas, numpy, scikit-learn 1.x, joblib). Each one is saved as `ml/artifacts/<version>/model.joblib` plus a `metadata.json` holding features, split, metrics and limits. Every number in this section is copied from those files. **None of them creates an alert, decides an emergency, or touches medication.**

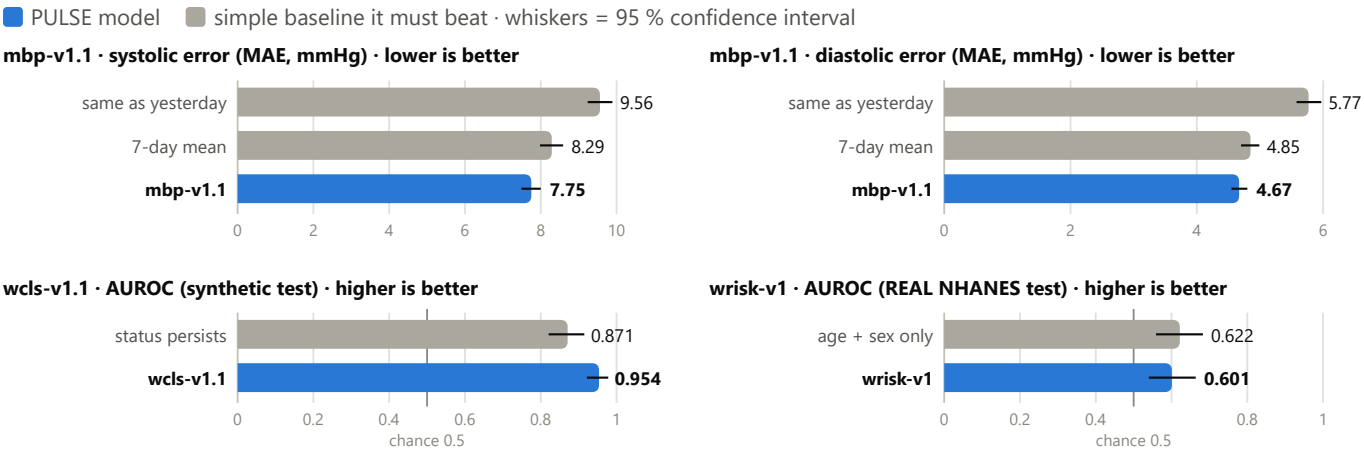


Figure 3. Each model against the simple method it has to beat, on held-out test data. mbp-v1.1 and wcls-v1.1 beat their baselines with confidence intervals that don't overlap. On real people, wrisk-v1 is no better than age and sex, so it isn't shown.

6.1 How every model is trained and tested (same recipe)

1. Data
Generate the labelled synthetic cohort, or load real data from `data/external/`

2. Features
Same code in training and serving (`features.py`, `ring.night_table`, `week_examples`)

3. Split by patient
70 / 15 / 15, seed 7; no patient in two sets; features only from earlier days

4. Candidates
A simple model (ridge or logistic) vs gradient boosting

5. Choose on validation
Pick the best on validation patients only

6. Test once
On unseen test patients, vs simple baselines; 95 % bootstrap CI over patients

7. Save
`model.joblib` + `metadata.json`, versioned folder

8. Serve
A batch job loads it and writes only its own table

Tests check the recipe: no patient overlap, no feature from the target day or later, deterministic retraining, and jobs that write only their own table.

6.2 `mbp-v1.1`: next-day morning blood pressure SERVED

Use	Informational estimate in the **doctor's** view, labelled "Model estimate (synthetic training data)". Never shown to patients, never used for alerts or medication.
Target	Mean systolic and diastolic of the next patient-local day's readings before 12:00.
Inputs (21 features)	From valid readings of the 7 days before only: last morning value, 7-day and 3-day means, 7-day morning mean, 7-day slopes, systolic spread, morning−evening gap, reading count, days with readings, day of week (7 flags).
Eligibility	≥ 6 valid readings on ≥ 4 local days in the 7 days before, and ≥ 1 morning reading; otherwise `insufficient_history` (no numbers).
Training data	`pulse-sim-v2`, 200 patients × 90 days (36,890 readings, seed 7).
Split	By patient: 140 / 30 / 30 patients → 8,730 / 1,918 / 1,929 patient-days.
Model choice	Validation MAE (mean of sys and dia): **ridge 6.18** vs gradient boosting 6.30 → ridge.
Uncertainty	Split-conformal interval from validation patients: ±16.2 systolic / ±9.6 diastolic; on test it covered the true value **90.1 % / 89.5 %** of the time (target 90 %).

Test: 30 unseen patients, 1,929 days	systolic MAE (95 % CI)	systolic RMSE	diastolic MAE (95 % CI)	diastolic RMSE
"same as yesterday's morning"	9.56 (9.24–9.89)	12.04	5.77 (5.58–5.97)	7.31
"7-day mean"	8.29 (7.98–8.59)	10.42	4.85 (4.70–4.99)	6.11
mbp-v1.1	**7.75 (7.49–8.00)**	**9.80**	**4.67 (4.55–4.80)**	**5.87**

Reading: about 7.8 mmHg off on systolic, better than both simple methods. The error is larger than the first version's 5.2 because the training data is now as variable as real home BP; this is the more honest number. **Real-data check:** the only open real home series

has 1 reading every 3 days, so the model can't run on it; it was used to calibrate the simulator instead.

6.3 `ring-v1.1`: ring monitoring FLAGS ONLY

Use	Advisory flags for the doctor, never an alert, never the safety path.
Night statistic	Median of each metric 00:00–06:00 local (HR, HRV, SpO2, skin temperature), built from raw `wearable_samples` by the same code in training and serving.
Flag 1: reference	SpO2 night median < 94 % (BTS; **88 % with COPD**) or resting HR > 100 bpm (Mayo). Live 10-minute windows need ≥ 2 samples.
Flag 2: personal	Doctor-approved band if one exists; else robust z = (value − median) / (1.4826 × MAD) over the 28 nights before (≥ 14 nights, confirmed anomalies left out), flag if |z| > **3.5**.
Flag 3: multivariate	IsolationForest on the 4 personal z-scores (contamination 0.01, a synthetic setting). Skipped with atrial fibrillation; HRV is never flagged with AF.
Training data	Synthetic ring data for the same 200 × 90 cohort (1,005,144 samples), 67 injected episodes. Split by patient 140 / 30 / 30, seed 7.

Method (synthetic test, 30 patients)	episodes found	flagged nights	night precision	false flags / patient-week
reference only	2 / 7	140	0.029	0.39
personal baseline, z > 3.5	**3 / 7**	104	0.077	0.27
IsolationForest	1 / 7	23	0.130	0.06
personal or IsolationForest	3 / 7	107	0.075	0.28

z limit (validation only)	episodes found	false / patient-week
2.5	9 / 10	0.91
3.0	7 / 10	0.44
3.5 (kept)	**7 / 10**	**0.22**

Why 3.5: same recall as 3.0 with half the false flags. 2.5 would mean about one false flag per patient every week. The doctor's band makes the limit patient-specific. 7 and 10 episodes are small samples.

Real-data test: PhysioNet Apnea-ECG (8 real nights)

Recording	Group	hours	SpO2 median	minimum	% time < 90	ODI-4 / h	ring-cadence median (6 samples)	Ring rule flags?
a01r	apnea	8.2	91	66	45.3	70.5	82.0	**yes**
a02r	apnea	8.8	86	61	61.0	56.5	83.5	**yes**
a03r	apnea	8.7	92	50	26.1	57.7	87.5	**yes**
a04r	apnea	7.5	86	50	61.1	59.4	83.5	**yes**
b01r	borderline	8.0	96	51	0.5	1.5	96.0	no
c01r	control	7.8	96	57.5	0.1	0.3	96.0	no
c02r	control	8.1	95	62.5	0.3	2.1	95.0	no
c03r	control	7.6	95	69.5	0.1	0.4	96.0	no

Reading: the night-median rule flagged **4 / 4** apnea nights and **0 / 4** borderline / control nights. A night-**minimum** rule would fail, because even controls hit 57–69 % from sensor artefacts. This is a sleep-lab oximeter and only 8 nights: a sanity check, not a validation. The simulator's sleep apnea is milder than these severe cases (real ~60 dips per hour).

6.4 `wcls-v1.1`: will next week be above target? IN THE WEEKLY REPORT

Label	Next week's mean of valid home readings ≥ the patient's target (systolic OR diastolic). Synthetic targets 135/85 or 130/80; default 135/85.
Inputs (11 features)	This week's mean systolic and diastolic, reading count, morning and evening systolic, slope, spread, last-3 systolic, gaps to target (sys, dia), "above target now".
Split	By patient, seed 7: 1,791 / 384 / 387 patient-weeks.
Model choice	Validation AUROC: **logistic regression, BP only 0.959**; gradient boosting BP only 0.941; logistic + ring 0.956; boosting + ring 0.944. Ring features didn't help, so BP only.

Test (387 weeks, 240 above target)	AUROC (95 % CI)	balanced accuracy	Brier (lower = better)	Confusion [TN FP / FN TP]
wcls-v1.1	**0.954 (0.922–0.978)**	0.892	**0.080**	130 17 / 24 216
"status persists"	0.871 (0.821–0.915)	0.871	0.124	125 22 / 26 214
only the 48 weeks where status changes	0.126 (0.054–0.197)	0.210	0.508	5 17 / 21 5

Reading: better ranked and better calibrated than "nothing changes", but it has no skill on the weeks that actually change. It mostly says "same as now". So the report shows it as **"how close to the target line"**, not as a predictor of change.

6.5 `mbp-v2.1`: does ring data help predict BP? EXPERIMENT

Same split and recipe as `mbp-v1.1`, plus 7 ring features (last night's HR, HRV, SpO2, temperature; 7-night HR and HRV; ring nights in 7). Validation ridge 6.125 vs 6.178.

Test MAE (synthetic)	systolic	diastolic
v1.1 features	7.75 (7.49–8.00)	4.67 (4.55–4.80)
+ ring night features	7.71 (7.45–7.97)	4.64 (4.51–4.77)

Decision: the 0.04 mmHg gain is within the noise, and the ring-to-BP link in the simulator is invented, so any gain is built in. `mbp-v1.1` stays served.

6.6 `wrisk-v1`: trained on REAL people (NHANES) NOT SHOWN

Question	Is a treated adult's exam BP uncontrolled (mean ≥ 140/90)?
Population	US adults 18+ taking prescribed BP medicine, ≥ 2 valid exam readings: 2,309 people. Survey weights used as sample weights.
Features (12, only fields PULSE stores)	age, female, BMI, waist, current / former smoker, diabetes, high cholesterol, kidney disease, COPD, heart failure, stroke.
Split	By person, seed 7: 1,616 / 346 / 347. Weighted uncontrolled rate in test: 30.9 %.
Model choice	Weighted validation AUROC: logistic 0.568 vs gradient boosting 0.511 → logistic.

Test (347 people)	AUROC (95 % CI)	weighted	bal. acc.
wrisk-v1	0.601 (0.540–0.664)	0.623	0.528
age + sex only	0.622 (0.559–0.683)	0.611	0.500

Calibration (test quintiles)	predicted	observed
1 (lowest)	0.22	0.31
2	0.29	0.26
3	0.34	0.39
4	0.38	0.41
5 (highest)	0.46	0.47

> Odds ratios per unit: female 1.38, stroke 1.37, kidney disease 1.32, diabetes 1.29, current smoker 1.23, age 1.016 per year, high cholesterol 0.76, COPD 0.62. With the 130/80 label: weighted AUROC 0.624. **Reading:** on real data, these risk factors can't tell whose BP is controlled under treatment; control has to be **measured**, which is exactly what PULSE does. A near-random score would mislead a doctor, so it isn't displayed. Limits: US population, exam BP, self-reported diagnoses, one visit.

6.7 Real population "safe band" for the cuff pulse (NHANES)

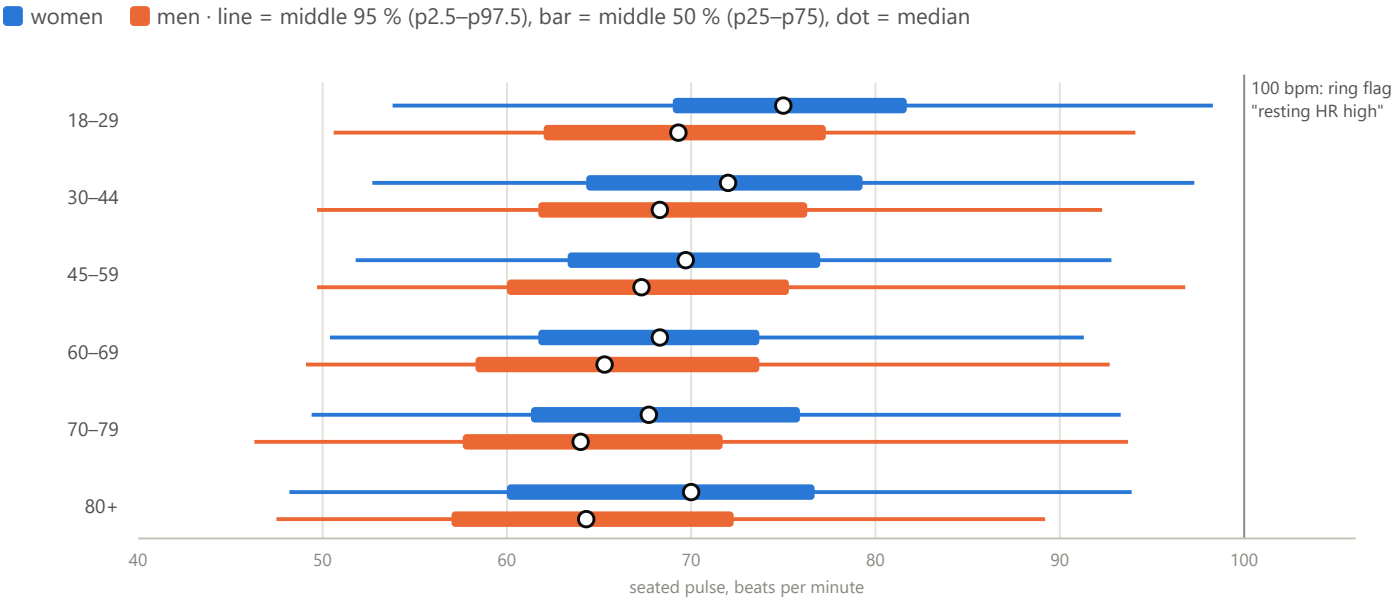


Figure 4. Real seated pulse of 7,296 US adults (NHANES 2017–2020, survey-weighted, mean of ≥ 2 valid readings). The middle 95 % is about 46–98 bpm in every group, so a pulse under 60 is common. That's why ring rules flag only a **high** resting HR.

How PULSE uses it: the weekly report places the patient's 7-day mean **seated cuff** pulse as a percentile among US adults of the same sex and age (`summary.pulse_vs_population`). It's context, not a threshold, and it is never applied to ring night HR (a different measurement). BP percentiles are also stored, but home BP categories use the ACC/AHA home column, because NHANES BP is exam BP.

6.8 Medical reference values (reviewed by the team's medical mentor, 30 Oct)

Value	Used where	Source
Home BP categories: normal, elevated 120, stage 1 130/80, stage 2 **135/85** (= office 140/90)	weekly report `home_bp_category` (OR rule on the week's home mean)	ACC/AHA 2017, Table 11, home (HBPM) column
SpO2 < 94 % (COPD < 88 %)	ring flag (reference), band floor, R-RING-1	BTS oxygen guideline 2017
Resting HR > 100 bpm (high side only)	ring flag (reference), band ceiling	Mayo Clinic (adults at rest 60–100)
HRV, skin temperature	personal baseline only; HRV ignored with AF	no population reference used
Oximeter accuracy limits	wording, limits text	FDA safety communication, 19 Feb 2021

Proposed safety rule R-RING-1, for the backend: SpO2 median over 10 minutes < 94 % with ≥ 2 samples → prompt a cuff reading, a symptom check and re-seating the ring. Never an emergency by itself; symptoms keep their own emergency path. It runs on raw `wearable_samples` in the backend, so the safety path never waits for an ML job. Spec: `ml/pulse_ml/reference.py` (`R_RING_1`). Reviewed by the medical mentor on 30 Oct. **TO BUILD** (backend)

7. Live ring monitoring and the weekly doctor loop

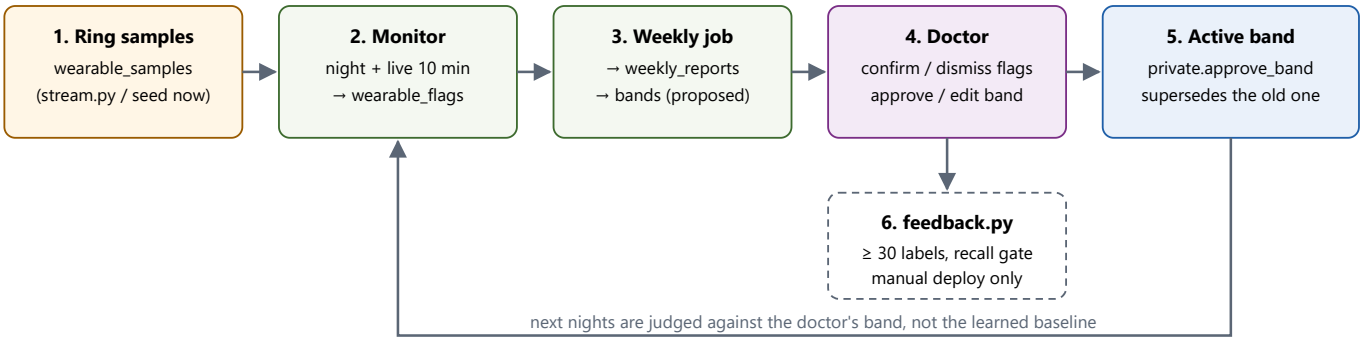


Figure 5. The weekly loop. The model only proposes; a doctor activates a band; labels can retune personal thresholds, never the medical references, and never deploy themselves.

Piece	Rule
Monitor, night mode	`monitor --nights`: every assigned patient's nights; reference, then band or personal z, then IsolationForest. One flag per (patient, metric, kind, window), so re-runs never duplicate.
Monitor, live mode	`monitor --follow --every 5`: polls new samples by `created_at` with a 5-minute overlap (catches late rows); 10-minute windows; band flags need 2 windows in a row.
Weekly report	Patient-local Monday–Sunday. `home_bp_category` from valid home readings; `control_status` vs the active care-plan target (default 140/90; insufficient_data under 3 readings); `wcls-v1.1` probability when ≥ 3 valid readings; ring night summary, flag counts, latest BP estimate, pulse placement (NHANES). A report the doctor already reviewed is never overwritten.
Band proposal	Median of the last 28 nights ± max(3 × 1.4826 × MAD, floor). Floors: HR 5 bpm, HRV 5 ms, SpO2 1.5 %, temperature 0.5 °C. Clipped to the reference on the safety side (SpO2 low ≥ 94, or ≥ 88 with COPD; HR high ≤ 100); SpO2 capped at 100. Needs ≥ 14 nights; none for HRV with AF.
Feedback (review M5)	Doctor-reviewed flags become labels; ≥ 30 needed; they retune only `personal_anomaly` thresholds; a retrained model must not lose injected-episode recall (`feedback --gate`); no automatic deployment.

Verified on the local copy (30 Oct 2026, all 12 demo patients with a simulated ring)

- `monitor --nights`: 34 flags. Karim's seeded HR rise is caught on its 2 nights. Karim and Samir (sleep apnea) get low night SpO2 flags. Fatima (COPD, night SpO2 ~92 %) gets **no** false SpO2 flag thanks to the 88 % limit; before the COPD rule she had 5. Houria (AF) gets no HRV flag.
- Live: `stream --scenario spo2_dip` + `monitor --follow` gives Nadia a live SpO2 flag within about 10 seconds.
- `weekly`: 11 reports, 43 band proposals (e.g. Fatima's SpO2 band 89.8–92.8 %, waiting for the doctor). A test approves and edits Karim's HR band through RLS, re-runs the monitor, and checks that the flags now cite `band:<id>`.

8. How the jobs run (inference contract)

Job	Reads	Writes	Rules
`predict`	Valid readings of actively assigned patients, strictly before the target day in the reading's own time zone	One row per patient, target day, model version in `ml_predictions` (upsert)	Default target = tomorrow (patient-local). Values and intervals null unless `ok`. `trained_on` = `synthetic:pulse-sim-v2`. Unassigned patients (Yacine) get no row.
`monitor`	`wearable_samples`, active bands, condition codes, confirmed flags	`wearable_flags` only	Advisory. Never alerts.
`weekly`	Readings, care plans, ring nights, flags, predictions, profile + body measurements	`weekly_reports`, proposed `patient_bands`	Numbers-only summary; reviewed reports are kept.

Backend mapping: endpoint 24 reads the latest prediction for (patient, date), newest model first; no row → `model_unavailable`. Still to build by the backend: weekly report endpoint, flag review, band approval through `private.approve_band`, R-RING-1. All jobs print counts only, never health values.

9. Tests and quality

File	tests	What it guards
`tests/test_db.py`	48	Validation policy cases; duplicates flagged not dropped; labelling constraints; import dedup and same-file refusal; one active care plan; patient / doctor / admin / logged-out visibility; forbidden writes; doctor notes hidden; ML tables doctor-only; Algerian fields; wearable checks; band approval supersedes and is audited; live REST API enforces access.
`tests/test_ingest.py`	5	CSV parsing, row-level rejections, missing columns, unknown time zone, deterministic seed labelled simulated.
`tests/test_simulate_stability.py`	5	SHA-256 fingerprints of the BP data (ring and condition changes can't alter BP); ring samples pass the table checks; the cloud refresh deletes only simulated demo-ring samples; chronic conditions shape ring values.
`ml/tests/test_ml.py`	6	Home-BP categories; no patient overlap; features only from earlier days; short history → insufficient; deterministic training; the predict job writes only predictions for assigned patients.
`ml/tests/test_ring.py`	7	CSV and database give identical nights; conditions change references; baseline uses earlier nights and skips confirmed anomalies; doctor band overrides baseline; live windows; late rows caught; seeded episode flagged.
`ml/tests/test_weekly.py`	3	Band maths; regression gate blocks lower recall; full weekly report and doctor band loop through RLS.
`ml/tests/test_classify.py`	3	Next-week label with the OR rule; features never see the label; deterministic training; the report gets a probability.
`ml/tests/test_real.py`	9	Raw real data is git-ignored; weighted percentiles; NHANES mapping and population; PULSE fields map to the same features; Mendeley parsing; pulse placement band; ODI-4 counts real desaturations.

Latest full run (30 Oct 2026): 86 / 86 passed on the local database. The full suite gave 84 passed, 2 skipped (the live-login tests need the demo password in the environment); those 2 then passed when run with the password loaded from `.env`.

10. Commands (Windows, repo `Desktop\pulse`)

Command	What it does
`npx -y supabase@2.122.0 start` / `db reset --local`	Local database (ports 55321–55323), all migrations + seed
`.venv\Scripts\python -m pulse_data.simulate seed`	Write the demo seed (password from `.env`)
`.venv\Scripts\python -m pulse_data.simulate csv`	ML cohort to `data/synthetic/` (git-ignored)
`cd ml` then `..\.venv\Scripts\python -m pulse_ml.train`	Train + test `mbp-v1.1` (~70 s)
`... -m pulse_ml.ring_train` · `classify` · `mbp_v2`	Train + test `ring-v1.1`, `wcls-v1.1`, `mbp-v2.1`
`... -m pulse_ml.real_data fetch all`	Download the 3 real datasets (~200 MB) + manifests
`... -m pulse_ml.real_bands` · `wrisk` · `real_tests all`	NHANES bands, `wrisk-v1`, real-data tests

`... -m pulse_ml.predict --all` · `monitor --nights` · `weekly`	The three jobs (local by default; `DATABASE_URL` for the cloud)
`... -m pulse_data.stream --scenario spo2_dip --minutes 1`	Simulated live ring (refuses the cloud without `--i-confirmed-cloud`)
`scripts\cloud_update.ps1`	Cloud: pending migrations + ring demo refresh + check (DONE 30 Oct)
`scripts\run_cloud_jobs.ps1`	Cloud: run the 3 jobs (`DATABASE_URL` from `.env`, or a temporary login from the linked Supabase CLI) (DONE 30 Oct)
`.venv\Scripts\python -m pytest`	All tests (local stack running)

11. Safety, ethics and limits

- **ML is advisory only.** No model creates alerts, decides an emergency, or suggests starting, stopping or changing medication. Symptoms trigger the emergency path on their own.
- **Patients never see ML output** until the backend owner and a clinician decide what a patient should see.
- **Synthetic is always labelled.** Metrics on synthetic data show the pipeline works; they say nothing about real patients.
- **Not validated for Algeria.** NHANES is the US population; Mendeley is one patient; Apnea-ECG is 8 lab nights.
- **Data terms respected:** NHANES statistics only (no re-identification, no linkage); Mendeley non-commercial; PhysioNet attribution.
- **Before any real use:** real, consented daily home readings; retraining and evaluation by patient and over time; clinician review of thresholds, wording and display; drift monitoring.

12. Next steps

#	Step	Who	Status
1	Run the 3 ML jobs on the cloud to fill the 4 ML tables, then a read-only check	Data/AI	DONE 30 Oct
2	Share the demo password privately; send the backend the message in `docs/data/CLOUD_STEPS.md`	You	OPEN
3	Backend: alert rules writing to Supabase; weekly report, flag review and band approval endpoints; R-RING-1	Backend	TO BUILD
4	Frontend: doctor panels for the BP estimate, ring flags, weekly report, band approval; profile fields	Frontend	TO BUILD
5	Clinician review: 180/120, red flags, SpO2 94 / 88 %, band rule, wording	Medical mentor	DONE 30 Oct
6	Algerian emergency number: **14**. Backend sets `dz-14` to verified in `api/app/config.py`	Team, then backend	DECIDED 30 Oct
7	Choose a real device (validated upper-arm cuff recommended) and build its adapter	Team, then data/AI	MISSING

13. Sources

1. Moussouni A. et al. Prevalence and risk factors of prehypertension and hypertension in Algeria (STEPS 2016–2017). *BMC Public Health* 2022. pmc.ncbi.nlm.nih.gov/articles/PMC9386961
2. National Center for Health Statistics. NHANES 2017–March 2020 pre-pandemic data files (P_DEMO, P_BPXO, P_BMX, P_SMQ, P_DIQ, P_BPQ, P_TCHOL, P_MCQ, P_KIQ_U). Data user agreement: cdc.gov/nchs/data_access/restrictions.htm
3. Dvornik O., Chuiko G., Baganov Y., Shyian I. Trends and seasonality in home blood pressure monitoring. Mendeley Data v1, 2017. doi:10.17632/pjcv33hpp7.1 (CC BY-NC 3.0)
4. Penzel T., Moody G.B., Mark R.G., Goldberger A.L., Peter J.H. The Apnea-ECG Database. *Computers in Cardiology* 2000;27:255–258; Goldberger A.L. et al. PhysioNet. *Circulation* 2000. (ODC-By 1.0)
5. Whelton P.K. et al. 2017 ACC/AHA high blood pressure guideline, Table 11 (corresponding office and home values).
6. O'Driscoll B.R. et al. BTS guideline for oxygen use in adults. *Thorax* 2017;72 Suppl 1.
7. Mayo Clinic. Heart rate: what's normal? (adults at rest 60–100 bpm).

8. US FDA safety communication: pulse oximeter accuracy and limitations, 19 Feb 2021.
9. Quer G. et al. Inter- and intraindividual variability in daily resting heart rate (92,457 wearable users). *PLOS ONE* 2020.
10. Gries R.E., Brooks L.J. Normal oxyhemoglobin saturation during sleep. *Chest* 1996;110:1489–1492.
11. Tageldin M.A. et al. COPD prevalence in the Middle East and North Africa (BREATHE study), 2012.
12. HRV in type 2 diabetes: systematic review and meta-analysis. *PLoS ONE* 2018 (PMC5880391).
13. Obstructive sleep apnea and resistant hypertension, multicenter cohort of 284 patients (PMC6513347).
14. Perez M.V. et al. Apple Heart Study, ACC 2019 / *NEJM* 2019.
15. PULSE repository, branch `AI`: `ml/MODEL_CARD.md`, `ml/artifacts/*/metadata.json`, `docs/data/*.md`, `supabase/migrations/*`.

Alert thresholds, SpO2 limits, the band rule and wording were reviewed by the team's medical mentor on 30 Oct 2026 for prototype use; validation limits are engineering defaults. Study figures are quoted from the sources above.